COURSE DESCRIPTION FORM

INSTITUTION Bahria University

PROGRAM (S) TO BE EVALUATED BS (Computer Science)

A. Course Description

(Fill out the following table for each course in your computer science curriculum. A filled out form should not be more than 2-3 pages.)

Course Code	GSC – 122
Course Title	Probability & Statistics
Credit Hours	03
Prerequisites by Course(s) and Topics	Elementary concepts of mathematics
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Quizzes 10% Assignments 20% Mid Term Exam 20% Final Exam 50%
Course Coordinator	
URL (if any)	
Current Catalog Description	
Textbook (or **Laboratory Manual** for Laboratory Courses)	Introduction to Statistics by Ronald Walpole
Reference Material	***Reference Books:*** a) Statistics by Donald H.Sanders. b) *Introductory Statistics by Neil A. Weiss (7th Edition)* c) A first course in Statistics by James T.Mc Clave. d) General Statistics by Warren Chase/Fred Bown Business Statistics by Ken Black e) Basic Business Statistics by Mark L. Benenson f) Elementary statistics by Allen G. Bluman. (8th Edition)
Course Goals	➢ This course provides the basic methodology for estimation, hypothesis testing, and model fitting in variety of settings. ➢ Numerous examples help demonstrate fundamental concepts of statistical reasoning and research design.

	Coverage includes analytical techniques for predictor variables. While the procedures discussed are all immediately applicable, the wide spectrum of topics presented also provides a solid background for advanced training in specialized areas of statistics.
Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	

Week	Hours	Topics Covered
1	1 Hour	Introduction to course
	1 Hour	Basic definitions of descriptive and inferential statistics, Qualitative & Quantitative data
	1 Hour	Graphical representation, Stem & leaf plot, Box Whisker plots
2	1 Hour	Measures of central tendency(Introduction)
	1 Hour	Mean, Median & Mode, Quartile (ungroup data)
	1 Hour	Mean, Median & Mode, Quartile (Group data)
3	1 Hour	Variance , Standard Deviation, Coefficient of variation , Quartile Deviation, Coefficient of Q.D(ungroup data)
	1 Hour	Variance , Standard Deviation, Coefficient of variation , Quartile Deviation, Coefficient of Q.D(Group data)
	1 Hour	Practice session
4	1 Hour	Counting techniques, Permutation & Combination
	1 Hour	Probability Concepts: Sample Space, Events, & Null Space. Operations with Events: Intersection, Mutually Exclusive event Union, & Complement of an Event,
	1 Hour	Venn Diagrams, Problem Solving, Counting Sample points: Multiplication Rule. Definition & Basic properties of probability.
5	1 Hour	Probability of Mutually Exclusive Events, Probability of Not Mutually Exclusive Events.
	1 Hour	Probability of Independent & Dependent Events
	1 Hour	Application of randon variable, Special Probability Distributions: Binomial Probability Distributions
6	1 Hour	Practice session
	1 Hour	Poisson Probability Distribution
	1 Hour	Negative Binomial , Exponential & Gamma distribution
7	1 Hour	Normal Probability Distribution. Area under the Normal curve
	1 Hour	Linear Regression: Scatter Diagram & Regression Equation
	1 Hour	Linear Correlation: Coefficient of Correlation
8	1 Hour	Problem Solving, Strength of Coefficient of Correlation
	1 Hour	Coefficient of Determination: Probable Error
	1 Hour	Practice session
9	Mid Term	
10	1 Hour	Introduction (Sampling with & without replacement)
	1 Hour	Sampling mean and variance, standard error
	1 Hour	Central limit theorem

	11	1 Hour	Estimation
		1 Hour	Hypothesis: Introduction
		1 Hour	Null & Alternative Hypothesis
	12	1 Hour	Establishing the Hypothesis, Type-1 & Type-2 Error
		1 Hour	Level of Significance, Test Statistic, & Critical Region Level of Significance, Test
		1 Hour	One tailed & Two tailed tests
	13	1 Hour	Rejection Rule & Conclusion
		1 Hour	Examples
		1 Hour	Testing the Population Mean by *Z-test & t-test*
	14	1 Hour	Problem Solving
		1 Hour	Problem Solving
		1 Hour	Problem Solving
	15	1 Hour	Testing the difference between Two Population Means by *Z-test*
		1 Hour	Problem Solving
		1 Hour	Problem Solving
	16	1 Hour	Testing the difference between Two Population Means by *t-test*
		1 Hour	Problem Solving
		1 Hour	Problem Solving
	17	1 Hour	Revision
		1 Hour	Revision
		1 Hour	Revision
	18	Final Exam	

Laboratory Projects/Experiments Done in the Course				
Programming Assignments Done in the Course				
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	16	16	16	0
Oral and Written Communications	Every student is required to submit at least 2 assignments and presentations of typically 20 minute's duration. Include only material that is graded for grammar, spelling, style, and so forth, as well as for technical content, completeness, and accuracy.			

Instructor Name:

Instructor Signature ________________

Date ________________